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JEL Classification: D72, R11

Keywords: Regional favoritism

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Regional favoritism: New data, larger sample, same pattern*

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February 2026

Abstract

Hodler and Raschky (2014) provide evidence for regional favoritism by documenting that subnational regions have more intense nighttime lights when they are the birth region of the country's current political leader than at other times. In this paper, we test the robustness of their findings using new data on nighttime light emissions and the birthplaces of political leaders, resulting in a larger sample with many more countries and years. We confirm that leader birth regions have more intense nighttime lights and that this effect is larger in countries with high ethnic fractionalization, undemocratic institutions, and low levels of education.

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1 Introduction

Many political leaders choose policies that benefit the regions they come from. Possible reasons are that they assign higher welfare weights to people living in “their” regions, or that the support of these people is crucial for their political survival (as in [Padró i Miquel, 2007](#)). Ample anecdotal evidence documents such regional favoritism (see, e.g., [Posner, 2005](#), on Zambia). Early quantitative studies typically focus on a few specific policy dimensions in relatively few countries ([Franck and Rainer, 2012](#); [Kramon and Posner, 2013](#); [Burgess et al., 2015](#)).¹ [Hodler and Raschky \(2014, henceforth HR\)](#) introduce a different approach. They study regional favoritism by examining whether subnational regions experience more intense nighttime lights when they are the birth region of the country’s current political leader than at other times. Their underlying argument is that nighttime lights provide a broad measure capturing the aggregate distributive effect of many different policies. They rely on nighttime lights data from 1992–2009 and around 38,000 subnational regions from 126 countries. They find that nighttime lights are more intense in the leader birth regions. They show that this effect is larger in countries with high ethnic fractionalization, undemocratic institutions, and low levels of education.

Over the last ten years, scholars have applied many variants of HR’s approach. [De Luca et al. \(2018\)](#) and [Dickens \(2018\)](#) change the relevant spatial units from administrative regions to ethnic homelands to provide evidence of ethnic favoritism. Others change the focus from the country’s effective political leaders to other key figures. [Asatryan et al. \(2023\)](#) and [Cruzatti et al. \(2024\)](#) show that ministers of national cabinets and leaders of national parliaments also engage in regional favoritism. [Do et al. \(2017\)](#) uncover hometown favoritism among lower-ranked politicians in Vietnam, and [Khalil et al. \(2021\)](#) show that governors of Indian states engage in favoritism too. [Bomprezzi et al. \(2025\)](#) find that nighttime lights are also more intense in the birth regions of the political leaders’ spouses. In addition, many studies combine information about the birth regions of political leaders (or other key figures) with georeferenced data other than nighttime lights to study regional favoritism in specific domains, such as foreign aid allocation ([Dreher et al., 2019](#); [Bommer et al., 2022](#)), public health ([Widmer and Zurlinden, 2022](#)), or the promotion of peace ([Kammerlander and Unfried, 2025](#)).

This paper revisits HR’s original approach and tests whether their results still hold when using the novel nighttime lights datasets by [Li et al. \(2020\)](#), [Nechaev et al. \(2021\)](#), [Chen et al. \(2024\)](#), and [Chiovelli et al. \(2026\)](#), as well as the novel dataset on the birthplaces of political leaders by [Bomprezzi et al. \(2025\)](#). These new datasets allow us to

¹In pioneering work, [Franck and Rainer \(2012\)](#) rely on DHS data from 18 African countries to show that children are less likely to die as infants and more likely to attend primary school if they and their mother identify with the same ethnic group as the country’s president. Also relying on DHS data, [Kramon and Posner \(2013\)](#) document that identifying with the same ethnic group as the political leader has differential effects across five policy dimensions and three African countries. [Burgess et al. \(2015\)](#) document ethno-regional favoritism in road building in Kenya.

extend the sample period to 2023 and broaden the country coverage, thereby doubling the number of country-years included in the analysis. Our analysis confirms HR’s main findings: Subnational regions experience more intense nighttime lights when the current political leader is born there than at other times. This pattern is more pronounced in countries with high ethnic diversity, undemocratic institutions, and low levels of schooling.

Section 2 presents the data, Section 3 the empirical specification, Section 4 the results, and Section 5 a brief conclusion.

2 Data

The units of observation are subnational regions from all over the world. We focus on administrative regions at the second subnational level, such as prefectures in China, districts in India, or counties in the United States, which we identify using the [Global Administrative Areas \(2022\)](#) database (GADM 3.6). Our analysis is based on 45,489 regions from 147 countries and, in most specifications, the years from 1992–2023. Compared with HR’s original analysis, we include 21 additional countries and 12 additional years, thereby more than doubling the number of country-years.²

2.1 Nighttime lights

Nighttime lights have become a commonly used proxy for local economic development ([Donaldson and Storeygard, 2016](#)). Indeed, there is evidence that nighttime light intensity is highly correlated with GDP at the level of countries ([Chen and Nordhaus, 2011](#); [Henderson et al., 2012](#)) and subnational regions (HR, appendix B). Moreover, nighttime light intensity is also positively correlated with various measures of human development at the very local level ([Bruederle and Hodler, 2018](#); [Chiovelli et al., 2026](#)).

There are two original global nighttime lights datasets covering multiple years. First, the U.S. Air Force Defense Meteorological Satellite Program (DMSP) provides data from 1992–2013 based on its Operational Linescan System (OLS) ([Elvidge et al., 1997](#)). The DMSP-OLS data have a spatial resolution of 30 arc seconds, implying pixels of less than 1 km², and measure luminosity on a range from 0–63. The DMSP-OLS data suffer from difficulties in detecting low light emissions ([Määttä and Lessmann, 2019](#)), from top-coding ([Bluhm and Krause, 2022](#)), and from blooming ([Abrahams et al., 2018](#)). HR and other early studies that use nighttime lights as a proxy for economic development are all based on DMSP-OLS data (e.g., [Henderson et al., 2012](#); [Michalopoulos and Papaioannou, 2013](#)).

A second global nighttime lights dataset is collected by the Visible and Infrared Imaging Suite (VIIRS) on board the Joint Polar-Orbiting Satellite System and is available from

²Our broader cross-sectional coverage is due to the availability of more complete border data for ADM2 regions and to HR’s restriction to countries with at least half a million inhabitants.

2012 onward (Elvidge et al., 2017). The VIIRS data have a wider range of luminosity values, a higher spatial resolution, a better low-light detection, no topcoding, and less blooming than the DMSP-OLS data (e.g., Chen et al., 2024; Chiovelli et al., 2026).

Various research teams have leveraged the temporal overlap between the DMSP-OLS and the VIIRS data to harmonize these two original nighttime lights datasets. Harmonized datasets covering a longer time period are particularly valuable for applications that exploit intertemporal variation in nighttime lights, such as the HR approach to measuring regional favoritism. Li et al. (2020) and Nechaev et al. (2021) harmonize the two datasets by simulating DMSP-OLS-like luminosity values for the VIIRS data. In contrast, Chen et al. (2024) simulate VIIRS-like luminosity values for the DMSP-OLS data. Finally, Chiovelli et al. (2026) first correct the DMSP-OLS data for top-coding and blooming and then harmonize the two original nighttime lights datasets by simulating *corrected*-DMSP-OLS-like luminosity values for the VIIRS data. Their approach not only extends the time coverage of the DMSP-OLS data but also addresses some of its well-established weaknesses.

We use five nighttime lights datasets. The original DMSP-OLS data and the harmonized datasets by Li et al. (2020), Nechaev et al. (2021), Chen et al. (2024), and Chiovelli et al. (2026).³ The DMSP-OLS data range from 1992–2013, the dataset by Nechaev et al. (2021) from 1992–2019 (once complemented with DMSP-OLS data for years up to 2012), and the other three datasets from 1992–2023.

We compute $Light_{ict}(\psi)$ as the average luminosity of the pixels in region i in country c in year t according to dataset $\psi \in \{OLS, Nechaev, Li, Chen, Chiovelli\}$, where we label each harmonized dataset by the first author’s name for simplicity. By design, $Light_{ict}(OLS)$, $Light_{ict}(Nechaev)$, and $Light_{ict}(Li)$ are all highly correlated (with correlation coefficients above 0.98). $Light_{ict}(Chen)$ and $Light_{ict}(Chiovelli)$ are less strongly correlated with each other and vis-à-vis the three other measures (with correlation coefficients from 0.49 to 0.70), which is not surprising given the different approaches used to harmonize the two original nighttime lights datasets.

Our dependent variables, $\ln Light_{ict}(\psi)$, are the logarithms of the average regional nighttime light intensities based on the different datasets ψ . We follow HR and others (e.g., Michalopoulos and Papaioannou, 2013) in adding a small constant (0.01) before taking logs to avoid “losing” unlit observations, but we also present robustness tests in which we omit such a constant.

2.2 Leader birth regions

The Political Leaders’ Affiliation Database (PLAD) by Bomprezzi et al. (2025) contains rich information on the effective political leaders of all countries from 1989–2023. Impor-

³For years with DMSP-OLS data from multiple satellites, we use the data from the newer satellite.

tantly, this information includes the leaders’ birthplaces, with varying geospatial precision. We include all political leaders for whom the ADM2 region of birth is known.⁴

Our main explanatory variable, $Leader_{ict}$, is equal to 1 if the political leader of country c in year t was born in region i , and equal to 0 if they were born in another region. There is exactly one leader birth region in 79.7% of the country-years, while there are multiple leader birth regions in 14.2%. In the remaining 6.1%, we cannot determine a leader birth region because the leaders’ birthplaces are unknown, known only with low geospatial precision, or abroad. In these cases, we set $Leader_{ict}$ to missing for all regions of the given country-year.⁵

2.3 Further data

To control for regional population dynamics, we compute the log of regional population, $\ln Pop_{ict}$, using the Global Human Development population data by [Carioli et al. \(2023\)](#).⁶

To explore effect heterogeneity, we use country-level variables in the spirit of HR. First, we measure diversity using ethnic fractionalization by [Alesina et al. \(2003\)](#). Second, we employ two measures for democratic institutions: the Polity2 score from the Polity V Project ([Marshall and Gurr, 2020](#)), which was already used by HR, and the Liberal Democracy index from the Varieties of Democracy (V-Dem) Project ([Coppedge et al., 2023](#)), which has greater cross-sectional and temporal coverage. Third, the years of schooling by [Barro and Lee \(2013\)](#).⁷ Lastly, the country-level GDP per capita (PPP, in constant 2021 international dollars) by the World Development Indicators ([World Bank, 2024](#)).⁸

[Table 1](#) presents summary statistics for the region- and country-level variables used in our analysis.

⁴HR relied on the list of political leaders in the Archigos database ([Goemans et al., 2009](#)) and collected information on these leaders’ birthplaces. [Bomprezzi et al. \(2025\)](#) also build on the Archigos database, but extend it beyond 2015. They geocode the leaders’ birthplaces using the approach outlined in [Strandow et al. \(2011\)](#).

⁵Results are very similar when setting $Leader_{ict} = 0$ (instead of setting it to missing) for all regions of country-years for which we cannot determine a leader birth region.

⁶The regional population data are available at five-year intervals. We linearly interpolate to obtain annual population estimates.

⁷The Barro-Lee data are available at five-year intervals up to 2015. We linearly interpolate between data points and extrapolate beyond 2015 using the 2010–2015 trend.

⁸We take all these country-level variables from the Quality of Government dataset ([Teorell et al., 2026](#)).

Table 1: Summary statistics

Variable	Observations	Mean	S.D.	Min.	Max.
Panel A: Summary statistics for region-level variables					
$\ln Light_{ict}(OLS)$	931,087	0.280	2.431	-4.605	4.143
$\ln Light_{ict}(Nechaev)$	1,184,829	0.229	2.424	-4.605	4.143
$\ln Light_{ict}(Li)$	1,357,958	0.469	2.383	-4.605	4.143
$\ln Light_{ict}(Chen)$	1,357,958	-2.233	2.165	-4.605	6.782
$\ln Light_{ict}(Chiovelli)$	1,357,958	-0.617	2.596	-4.605	7.586
$Leader_{ict}$	1,357,989	0.004	0.060	0.000	1.000
$\ln Pop_{ict}$	1,354,870	3.265	1.788	-6.908	10.397
Panel B: Summary statistics for country-level variables					
$Fractionalization_c$	4,588	0.461	0.259	0.002	0.930
$Polity_{ct}$	3,928	3.450	6.294	-10.000	10.000
$LibDemocracy_{ct}$	4,615	0.400	0.275	0.005	0.897
$Schooling_{ct}$	3,940	8.137	3.043	0.975	15.398
$\ln GDP_{ct}$	4,448	9.206	1.213	6.236	11.840

Notes: [Section 2](#) describes all variables and provides their sources.

3 Empirical specification

To study whether subnational regions have more intense nighttime light when they are the current political leader’s birth region, we estimate the following main specification:

$$\ln Light_{ict}(\psi) = \beta(\psi)Leader_{ict-1} + \gamma_i + \delta_{ct} + \epsilon_{ict} \quad (1)$$

Following HR, we use lagged values of $Leader_{ict}$ to account for likely delays between the central government’s decision on the allocation of public funds and their arrival in the chosen regions, and also between the arrival of these funds and their effects on consumption, investments, and, finally, nighttime lights. We expect coefficients $\beta(\psi)$ to be positive, indicating that regions have more intense nighttime lights when being a leader birth region than at other times. The region-fixed effects γ_i absorb all time-invariant regional characteristics, and the country-year-fixed effects δ_{ct} absorb all shocks and changes common to all regions of a given country. We cluster the standard errors ϵ_{ict} at the country level.⁹

We examine pre-trends to address the concern that unobserved changes within regions could simultaneously increase nighttime light emissions and the likelihood of becoming a leader birth region. For this purpose, we use the dummy variables $FutureLeader1_{ict}$, $FutureLeader2_{ict}$, and $FutureLeader3_{ict}$, which indicate regions that are not currently a leader birth region, but will become a leader birth region in exactly 1, 2, or 3 years. These variables are mutually exclusive in the sense that at most one of them can be equal

⁹HR clustered the standard errors at the level of political leaders, i.e., the set of country-years in which the country was governed by the same leader (or the same set of leaders). Clustering at the country level is more conservative.

to 1 in a given region and year. Similarly, we define the dummy variables $PastLeader1_{ict}$, $PastLeader2_{ict}$, and $PastLeader3_{ict}$, which indicate regions that are not currently a leader birth region, but used to be a leader birth region up until exactly 1, 2, or 3 years ago.

4 Results

Panel A of Table 2 presents the main results. Column (1) uses the full DMSP-OLS data, available from 1992–2013. The estimated coefficient is 0.044. In comparison, the coefficient estimate was 0.038 in HR, who covered only the 1992–2009 period and fewer countries.¹⁰ Columns (2) and (3) present results based on the harmonized nighttime lights datasets by Nechaev et al. (2021) and Li et al. (2020), who generate a longer time series by using the VIIRS data to generate DMSP-OLS-like data for the post-2013 period. The corresponding coefficient estimates are even somewhat larger, which is consistent with higher levels of regional favoritism in the post-2013 period. Column (4) uses the harmonized dataset by Chen et al. (2024), who simulate VIIRS-like data for the pre-2013 period. The coefficient estimate is again similar, but slightly smaller. Lastly, column (5) uses the harmonized dataset by Chiovelli et al. (2026), who first correct the DMSP-OLS data for top-coding and blooming and then use the VIIRS data to generate *corrected*-DMSP-OLS-like data for the post-2013 period. We estimate a coefficient of 0.054.¹¹ All these coefficient estimates are statistically significant at the 5% level. They are also economically meaningful. Given our log-level specification, a coefficient estimate of 0.054 implies that average nighttime light intensity is $e^{0.054} - 1 = 5.6\%$ higher when a given region is a leader birth region than at other times. Henderson et al. (2012) find a country-level elasticity of nighttime lights with respect to GDP of around 0.3. HR document a similar elasticity at the subnational level in their appendix B. Hence, the regional GDP may be 1.5–2% higher when a region is a leader birth region than at other times.

Panel B tests the robustness of our results when omitting the small constant before taking the log of average nighttime light intensity. This has the disadvantage of losing observations with no detected nighttime light, but the advantage of making coefficients easier to interpret. In particular, the above formula for computing elasticities is not exact when a small constant is added before taking logs (Chen and Roth, 2024). However, comparing panels A and B reveals that while individual coefficient estimates change, they remain similar on average (with the average coefficient being 0.048 in panel A and 0.046 in panel B).

Panel C tests the robustness of our results when controlling for regional population (in logs). The coefficient estimates of interest are quantitatively very similar to those in

¹⁰The coefficient estimate is even closer to the one of HR, namely 0.039, if we restrict the sample to 1992–2009 period, despite us relying on the PLAD data and including more countries.

¹¹Using the original HR dataset, Chiovelli et al. (2026) also show that the coefficient estimate gets larger when using their harmonized data instead of the DMSP-OLS data.

Table 2: Evidence for regional favoritism

	(1)	(2)	(3)	(4)	(5)
Dependent variable:	$\ln \text{Light}_{ict}(\psi)$				
Data source ψ :	OLS	Nechaev	Li	Chen	Chiovelli
Panel A: Main results					
Leader_{ict-1}	0.044** (0.018)	0.056*** (0.016)	0.048*** (0.017)	0.040** (0.017)	0.054*** (0.018)
Observations	931,013	1,184,829	1,357,958	1,357,958	1,357,958
Panel B: Extensive margin					
Leader_{ict-1}	0.028* (0.015)	0.042*** (0.014)	0.040** (0.016)	0.063* (0.038)	0.056** (0.022)
Observations	843,841	1,072,845	1,254,595	1,150,813	1,188,526
Panel C: Controlling for population					
Leader_{ict-1}	0.044** (0.018)	0.056*** (0.016)	0.046*** (0.017)	0.037** (0.016)	0.052*** (0.018)
$\ln \text{Pop}_{ict}$	-0.030 (0.059)	0.026 (0.047)	0.061 (0.058)	0.160** (0.064)	0.081 (0.063)
Observations	928,906	1,182,127	1,354,869	1,354,869	1,354,869
Panel D: Dynamics					
$\text{PastLeader1}_{ict-1}$	0.010 (0.021)	0.026 (0.019)	0.027 (0.024)	0.035* (0.018)	0.018 (0.022)
$\text{PastLeader2}_{ict-1}$	0.005 (0.021)	0.027 (0.024)	0.016 (0.026)	0.017 (0.027)	0.012 (0.023)
$\text{PastLeader3}_{ict-1}$	0.013 (0.023)	0.022 (0.022)	0.027 (0.023)	0.010 (0.018)	0.004 (0.019)
Leader_{ict-1}	0.035 (0.022)	0.057*** (0.016)	0.052*** (0.019)	0.042** (0.021)	0.056*** (0.020)
$\text{FutureLeader1}_{ict-1}$	-0.023 (0.025)	-0.008 (0.021)	0.010 (0.024)	0.017 (0.022)	0.016 (0.024)
$\text{FutureLeader2}_{ict-1}$	-0.001 (0.024)	0.003 (0.021)	0.003 (0.023)	0.018 (0.021)	0.024 (0.021)
$\text{FutureLeader3}_{ict-1}$	-0.002 (0.024)	0.025 (0.025)	-0.007 (0.023)	0.058*** (0.019)	0.023 (0.019)
Observations	822,598	1,065,875	1,146,634	1,146,634	1,146,634
Region-fixed effects	Yes	Yes	Yes	Yes	Yes
Country-year-fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables, $\ln \text{Lights}_{ict}(\psi)$, are the log of average nighttime light intensity, computed based on the different nighttime lights data sources ψ indicated at the top of each column and discussed in Section 2.1. We add a small constant (0.01) before taking logs in panels A, C, and D, but not in panel B. The main explanatory variable, Leader_{ict-1} , indicates birth regions of the country's current political leader. All specifications include region- and country-year-fixed effects, and those in panel C also log population ($\ln \text{Pop}_{ict}$). Panel D includes dummy variables $\text{FutureLeader}X_{ict-1}$, indicating regions that are not currently a leader birth region, but will become one in exactly X years; and dummy variables $\text{PastLeader}X_{ict-1}$ indicating regions that are not currently a leader birth region, but were one until exactly X years ago. Standard errors are clustered at the country level and reported in parentheses. ***/**/* indicates statistical significance at the 1/5/10 percent level.

panel A, suggesting that the changes in nighttime lights are not driven by population dynamics specific to leader birth regions.

Panel D examines pre-trends. The estimated coefficients on $FutureLeader1_{ict-1}$, $FutureLeader2_{ict-1}$, and $FutureLeader3_{ict-1}$ are statistically insignificant (with the exception of a single coefficient being significant at the 10% level) and typically small. These estimates are reassuring in that they suggest that nighttime lights do not become more intense already in the years directly before a region becomes a leader birth region.

Panel D also looks at the years after regions cease to be a leader birth region. The estimated coefficients on $PastLeader1_{i,ct-1}$, $PastLeader2_{i,ct-1}$, and $PastLeader3_{i,ct-1}$ are statistically insignificant (with one exception) and typically small. This pattern suggests that the nighttime light premium associated with being a leader birth region vanes quickly after leadership changes.

Table 3 presents the results from interacting the main explanatory variable $Leader_{ict-1}$ with z-standardized country-level measures of ethnic diversity, democratic institutions, schooling, and GDP.¹² We see that the birth leader region effects increase with ethnic fractionalization (panel A), but decrease with the quality of democratic institutions (as measured by the Polity score in Panel B and the Liberal Democracy index in panel C) and schooling (panel D). This pattern is consistent with HR’s findings. The birth leader region effect also tends to decrease with GDP per capita (panel E), but, unlike in HR, the coefficients on the corresponding interaction term are not statistically significant.

These effects are sizeable. To illustrate this, we focus on the estimates in column (5), which are based on the harmonized data by Chiovelli et al. (2026). The estimates in panel A suggest that being the leader birth region increases nighttime light intensity by $e^{0.06} - 1 = 6.2\%$ in countries with average ethnic fractionalization (0.46, similar to Oman or Ukraine), but by $e^{(0.06+0.062)} - 1 = 13.0\%$ in countries with one standard deviation higher ethnic fractionalization (0.72, similar to Ethiopia or Sudan). Panel C suggests that being the leader birth region increases nighttime light intensity by 6.4% in countries with average democratic quality (i.e., a Liberal Democracy index value of 0.40, similar to Albania or Mexico when averaged over the sample period), but by 11% in countries where democratic quality is one standard deviation worse (a Liberal Democracy index value of 0.13, similar to Afghanistan or Burundi when averaged over the sample period). Finally, panel D suggests that being the leader birth region increases nighttime light intensity by 6.4% in countries with average years of schooling (8.1 years, similar to China or Colombia when averaged over the sample period), but by 14.5% in countries with one standard deviation fewer years of schooling (5.1 years, similar to the DRC or Uganda when averaged over the sample period).

¹²We z-standardize these variables in the country-year panel rather than our region-year panel to avoid giving more weight to countries with more ADM2 regions.

Table 3: Heterogeneity in regional favoritism

	(1)	(2)	(3)	(4)	(5)
Dependent variable:	$\ln \text{Light}_{ict}(\psi)$				
Data source ψ :	OLS	Nechaev	Li	Chen	Chiovelli
Panel A: Ethnic fractionalization					
$Leader_{ict-1}$	0.051*** (0.019)	0.060*** (0.016)	0.052*** (0.017)	0.043** (0.016)	0.060*** (0.018)
$Leader_{ict-1} \times Fractionalization_c$	0.059*** (0.019)	0.042*** (0.015)	0.035** (0.017)	0.054*** (0.018)	0.062*** (0.019)
Observations	922,200	1,173,646	1,345,195	1,345,195	1,345,195
Panel B: Polity score					
$Leader_{ict-1}$	0.063*** (0.024)	0.071*** (0.019)	0.063*** (0.020)	0.027 (0.018)	0.055** (0.021)
$Leader_{ict-1} \times Polity_{ct-1}$	-0.063** (0.025)	-0.052** (0.022)	-0.049** (0.023)	-0.012 (0.019)	-0.027 (0.024)
Observations	920,672	1,172,738	1,179,034	1,179,034	1,179,034
Panel C: Liberal democracy index					
$Leader_{ict-1}$	0.057*** (0.022)	0.067*** (0.018)	0.055*** (0.020)	0.046*** (0.018)	0.062*** (0.020)
$Leader_{ict-1} \times LibDemocracy_{ct-1}$	-0.054*** (0.018)	-0.050*** (0.015)	-0.036* (0.019)	-0.031** (0.015)	-0.042** (0.018)
Observations	930,229	1,183,373	1,356,294	1,356,294	1,356,294
Panel D: Years of schooling					
$Leader_{ict-1}$	0.043** (0.017)	0.060*** (0.015)	0.046** (0.019)	0.041** (0.018)	0.062*** (0.019)
$Leader_{ict-1} \times Schooling_{ct-1}$	-0.053*** (0.018)	-0.043*** (0.015)	-0.052*** (0.019)	-0.046*** (0.016)	-0.073*** (0.018)
Observations	881,590	1,121,195	1,284,929	1,284,929	1,284,929
Panel E: GDP per capita					
$Leader_{ict-1}$	0.043** (0.019)	0.056*** (0.016)	0.053*** (0.018)	0.040** (0.017)	0.057*** (0.019)
$Leader_{ict-1} \times \ln GDP_{ct-1}$	-0.022 (0.028)	-0.024 (0.023)	-0.032 (0.024)	-0.033 (0.020)	-0.036 (0.023)
Observations	907,995	1,154,815	1,323,476	1,323,476	1,323,476
Region-fixed effects	Yes	Yes	Yes	Yes	Yes
Country-year-fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variables, $\ln \text{Lights}_{ict}(\psi)$, are the log of average nighttime light intensity (plus 0.01), computed based on the different nighttime lights data sources ψ indicated at the top of each column and discussed in Section 2.1. The main explanatory variable, $Leader_{ict-1}$, indicates birth regions of the country's current political leader. Each panel includes interaction terms between $Leader_{ict-1}$ and one of the following z-standardized country-level variables: Ethnic fractionalization by [Alesina et al. \(2003\)](#) in panel A, the Polity2 score from the Polity V Project in panel B, the Liberal Democracy index from the V-Dem Project in panel C, the years of schooling by [Barro and Lee \(2013\)](#) in panel D, and log GDP per capita (PPP) by the World Development Indicators in panel E. All specifications include region- and country-year-fixed effects. Standard errors are clustered at the country level and reported in parentheses. ***/**/* indicates statistical significance at the 1/5/10 percent level.

5 Conclusions

We use novel data on nighttime lights and leader birthplaces to test the robustness of HR's finding of regional favoritism in a panel covering more countries and years (thereby doubling the number of country-years). We confirm that subnational regions experience more intense nighttime lights when the current political leader is born there than at other times. We also confirm that this pattern is more pronounced in countries with high ethnic diversity, undemocratic institutions, and low levels of schooling.

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